



Water Meter Reading Recognition Based on Deep Learning

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ABSTRACT

The widespread and large-scale distribution of water meter users in China poses a significant challenge to the traditional manual meter reading method in most regions of the country. Rapid and accurate meter reading for a large number of water meters distributed in different areas has become an urgent problem for various water supply companies. With the rapid development of deep learning, computer vision and other fields, more and more industrial and commercial projects are starting to utilize machine learning-based computer vision technology to process water meter readings, which previously required manual identification by human eyes. This paper mainly studies the problems of text area extraction and reading recognition of the dial-type water meter in real scenarios. Based on deep learning methods, DBnet semantic segmentation algorithm is first used for text detection, followed by CRNN convolutional recurrent neural network for text recognition. Regarding the difficulty of recognizing half characters, this paper proposes an agent character scheme by readjusting the format of the training set, which significantly improves the accuracy of half character recognition. The model is trained and tested on a dataset of real-scenario dial-type water meter images, and the test results show that the algorithm network still has good accuracy and fast recognition speed in complex scenarios such as low light, reflection, partial occlusion, angle tilt, and half-characters.

CCS CONCEPTS

• **Computing methodologies** → Machine learning; Artificial intelligence; Computer vision; • **Networks** → Network algorithms.

KEYWORDS

deep learning, water meter reading recognition, differentiable binarization, convolutional recurrent neural network

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1 INTRODUCTION

With the rapid development of deep learning, computer vision, image processing and recognition in recent years, various industries have begun to utilize deep learning-based computer vision technology to deal with the data that used to require human eye recognition [1] [2], therefore, domestic and foreign deep learning-based water meter readings recognition has also been studied in depth and has achieved phased results, the basic idea is to install a low-cost camera on the original water meter to capture images of water meter readings at specific intervals without changing the original water meter equipment, and upload the collection results through the network to a cloud-based deep learning reading recognition system for recognition and processing. This not only avoids significant and widespread renovation of the original water meter but also improves the automation, timeliness, and accuracy of water meter reading.

This paper aims to improve the basic methods for water meter reading detection and recognition based on deep learning by optimizing several issues related to performance and efficiency on the basis of reviewing the development history of water meter reading recognition research based on deep learning at home and abroad: Detection and localization segmentation of water meter readings in complex scenes; Pre-processing of text area under complex optical environment; The low efficiency of the overall detection and recognition process of the system and the low accuracy of text region recognition; The problem of half-character text recognition deviation.

This paper uses the Pytorch+Cuda deep learning framework as a foundation, with RNN and Differentiable Binarization as core algorithms. It combines text detection (semantic segmentation) and text recognition to detect the character regions of water meters and recognize their contents. The solutions to the above issues are attempted to improve the quality and accuracy of text region localization and text content recognition.

2 RELATED WORK

Character recognition in the area of water meter reading based on deep learning principles belongs to Optical Character Recognition (OCR). OCR technology was first proposed by German scientist Taussek in the 1920s, The initial application of OCR in water meter reading recognition is based on the traditional sliding template matching algorithm [3]. Wu Menglin from Nanjing University of Science and Technology proposed a sliding template matching algorithm to solve the case of differences in half-character templates, but this sliding template matching method has poor generalization ability for complex scenes [4]. To solve the problem of water meter digital area location in complex scenes, Li Y from Shenyang University of Technology introduced the Fourier transform reading region correction algorithm and Integral projection reading frame

addressing algorithm to improve the efficiency of water meter character frame localization, which provided the basis for many other reading region localization algorithms in the future [5]. Li Q's team from CCNU used the rectangular template matching technique in vertical order to recognize half-character data similar to the middle state of the word wheel by using a single rectangular box successfully [6], but this method is dependent on the cleanliness of the segmentation of the read region. Subsequently, feature recognition methods became popular and be widely used in the field of OCR, Liu Tianhua from Hunan University solved the problem of low recognition rate in the case of few character recognition samples by extracting structural features and stroke features from characters and performing classification operations using Support Vector Machine (SVM) algorithms to recognize the results [7]

Since 2014, the research results on scene text detection using deep learning algorithms have become increasingly abundant. Water meter text reading recognition scenarios can learn from the successful experiences of other scenarios. Dai Qing from Wuhan University of Technology utilized the special properties of character-based models and proposed a three-layer BP (Back Propagation) algorithm as a classifier for recognizing and extracting character-based models, achieving high recognition accuracy, but still faces challenges such as long training time and unstable network memory [8]. Nanjing Institute of Technology researchers, including Shuaicheng Pan, have proposed a novel character recognition method based on Convolutional Neural Network (CNN), addressing the low accuracy of traditional target matching algorithms. The approach primarily utilizes semantic segmentation principles for detection and typically generates feature maps from a full convolutional network (FCN). These feature maps are then subjected to post-processing steps to obtain character localization information, effectively enhancing the accuracy of water meter character recognition [9].

Zhang et al. from HUST use fully convolutional network to map character localization information for reasoning. They aggregate multiple character localization mapping information through MSER (Maximally Stable Extrernal Regions) and utilize the correctly localized character data trained on the dataset to exclude the remaining erroneous results [10]. In 2016, Liu W and others proposed a SSD (Single Shot MultiBox Detector) detection algorithm at the ECCV (European Conference on Computer Vision). This algorithm can infer character localization information and reasoning confidence probability of related results, which greatly improves the prediction efficiency and gives an intuitive representation of the data [11]. In 2018, at ICPR (International Conference on Pattern Recognition), Dai et al. from SJTU proposed a new text region segmentation model based on Resnet-101 residual network, combining the advantages of semantic segmentation model and target detection algorithms to realize the detection of the whole text region [12]. In the same year, Liu et al. from SenseTime published a paper in Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, proposing a unified end-to-end trained Fast Oriented Text Spotting With a Unified Network (FOTS). The RRI (Rotate the Region of Interest) attraction mechanism was used to merge the feature extraction process during both stages, greatly reducing the computational consumption. [13].

At the 2020 AAAI Conference (Association for the Advancement of Artificial Intelligence), Liao Minghui and others from the

Computer Vision Laboratory of Huawei proposed a natural scene text detection algorithm based on Differentiable Binarization. This algorithm can segment non-frontal scenes and various shaped text areas by accurately predicting pixel-level results[14]. In the process of text recognition, An End-to-End Trainable Neural Network for Image-based Sequ proposed by Shi and Bai from HUST has a stronger generalization ability and is more efficient in recognizing continuous text and text of different lengths[15].

3 WATER METER READING DETECTION AND RECOGNITION BASED ON DNNET AND CRNN

3.1 Differentiable Binarization

The current common practices for scene text localization fall into two main categories, one is based on regression-based models such as Textboxes++ and R2CNN, while the other relies on segmentation such as Pixel Connection Algorithm Networks, Instance Segmentation Networks, and Differentiable Binarization Networks. In real reading scenarios, text is presented in a variety of types, including slanted text, curved text, and parallel text. The segmentation-dependent approach is a text detection method capable of localization at the smallest level of the image, which allows more flexible and accurate character localization at different angles in a real water table environment.

In text localization algorithms that rely on segmentation [16], the generated probability map first uses Binarization to obtain foreground and background regions. The traditional method generally determines a threshold based on experience. The regions of the probability map that are greater than this threshold are considered as text regions, while those that are less than the threshold are classified as background regions. Finally, the binarized results can be further processed by certain post-processing algorithms to obtain the final text detection results. The DBnet introduced in this section incorporates a binarization operation in the training process of the network, which generates a threshold map for binarizing the generated probability maps. However, the traditional binarization operation is non-differentiable and cannot be trained by optimization methods such as gradient descent. Therefore, the researchers proposed a function similar to the binarization operation but differentiable as an alternative. This function is mathematically similar to the binarization operation, but can be optimized using gradient information during the training process, resulting in efficient training of threshold mappings.

$$B_{i,j} = \frac{1}{1+e^{-k(P_{i,j}-T_{i,j})}} \quad (1)$$

DBnet proposes a scheme which can globally reason about the threshold data of each pixel region in an image for bi-directional training using binarization operations in a segmented network, enabling the work of segmentation at the pixel level of a character region. Traditional segmentation-based target detection practices, as shown by the blue arrows in Figure 1, usually set an empirically determined threshold value which is used to output the resulting probability map as a binary map, then group the pixels into text instances by using some heuristic techniques such as pixel clustering; The practice of differentiable binarization grid is shown by

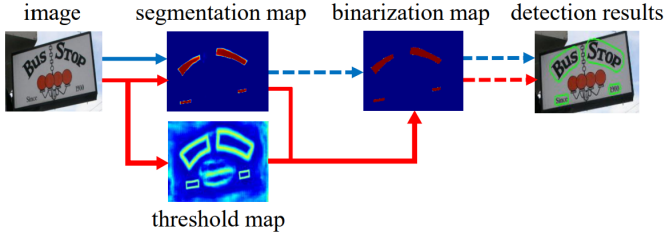


Figure 1: Comparison between traditional target detection methods and DBnet

the red arrow in the figure below, after obtaining the segmentation map, perform differentiable binarization with the threshold map generated by the network to obtain a binarized graph, then get the final image segmentation results after post-processing.

As shown in Figure 2, DBnet extracts features through the backbone network and then fuses them using a feature pyramid network structure similar to the Feature Pyramid Network. This results in two feature maps: a probability map and a threshold map. The two feature maps are then used to perform DB difference operations to obtain an approximate binary map for text regions. The approximate binary map is further processed to obtain information about the text regions.

The approximate binary map and the probability map train with the same set of annotated data. In actual prediction, during the actual prediction process, either the probability map or the approximate binary map can be generated to obtain the final prediction result. The results of these two methods are almost the same, but the DBnet author suggests using the probability map. This way, when making predictions, the DB module can be directly removed to reduce computation.

3.2 CRNN

In actual real-life scenarios, visual objects like handwriting and sheet music usually appear in sequences rather than separately and in isolation. The prediction object of this paper, water meter readings, also belongs to this class of scene text. Unlike traditional conventional target detection tasks, for such serialized targets, it is generally necessary to predict a series of target labels rather than just reasoning to derive individual data [17]. Two approaches are

commonly used to deal with the scene text recognition problem: one is to detect individual characters and then use a DCNN network to recognize the detected objects; the other is to view scene text recognition as a classification task, mapping a specific label for each sequential object. However, there are some limitations to these methods. First, in order to accurately detect and crop each character from an image, an accurate text recognition module is essential, which increases the complexity of the algorithm. Second, these methods are prone to difficulties and limitations when facing situations with rich combination variety such as musical scores and Chinese texts. Therefore, the existing methods may face dilemmas when dealing with text recognition problems in these complex scenarios.

As shown in Figure 3 Convolutional Recurrent Neural Network is mainly divided into Convolutional Layer (CNN), Recurrent Layer (RNN) and Translation Layer (CTC). The convolutional layer uses the Oxford University VGG (Visual Geometry Group) structure [18], which changes the kernel scale of the 3rd and 4th maxpooling from 2×2 to 1×2 . This transformation can be thought of as strip-cutting the image, where each stripe corresponds to a set of feature values and is arranged in a horizontal order, since text is ordered horizontally in the image. In order to mitigate the phenomenon of gradient vanishing that may occur during training, CRNN chose the Long Short-Term Memory (LSTM) neural unit [19] as a submodule of the recurrent neural network. It has been concluded that in sequence prediction tasks, both forward and backward information of sequences have a positive effect on the prediction performance. Therefore, to fully utilize the bidirectional information of sequences, CRNN employs a bidirectional RNN network [20]. The role of the translation layer in a CRNN model is to map the output of the recurrent layer to label sequences, which can be accomplished by finding the most probable label to complete the translation task. There are two modes for the translation layer: dictionary-based translation and non-dictionary-based translation. In the non-dictionary-based translation mode, the reasoning process is performed without any dictionary constraints, whereas in the dictionary-based translation mode, the output of the result is performed by selecting the list of labels with the highest likelihood.

3.3 Framework Design

As shown in Figure 4, the system first enhances the dataset by data augmentation to expand the sample data. The backbone of the

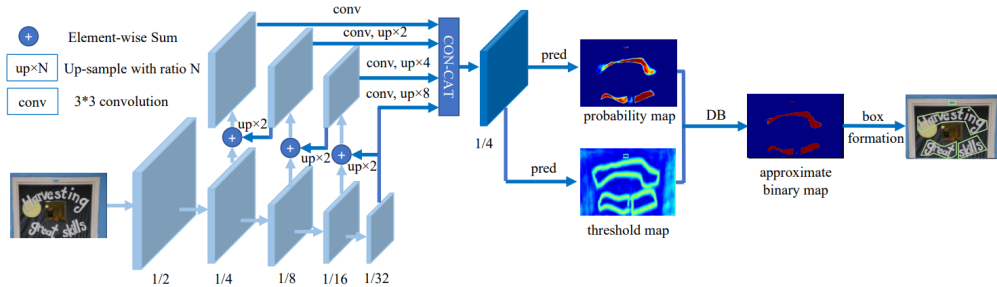
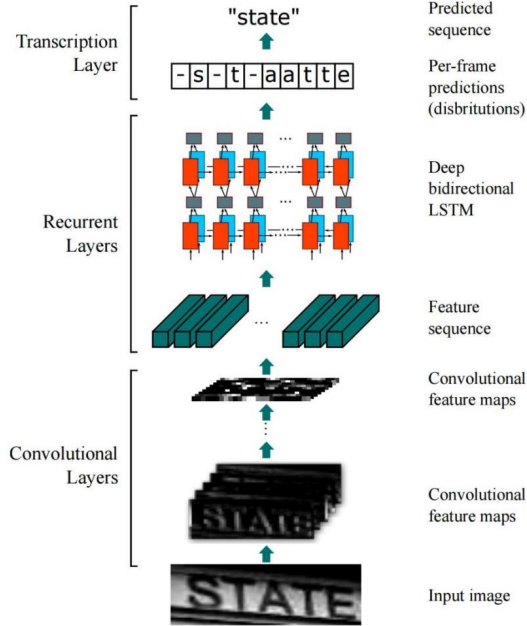


Figure 2: Process of DBnet to Acquire Textual Region Information

Table 1: VGG Convolutional Layer Structure

Type	Configuration
Transcription	-
Bidirectional-LSTM	#hidden units:256
Bidirectional-LSTM	#hidden units:256
Map-to-Sequence	-
Convolution	#maps:512,k:2 × 2,s:1,p:0
MaxPooling	Window:1 × 2, s:2
BatchNormalization	-
Convolution	#maps:512,k:3 × 3, s:1,p:1
BatchNormalization	-
Convolution	#maps:512,k:3 × 3, s:1,p:1
MaxPooling	Window:1 × 2, s:2
Convolution	#maps:256,k:3 × 3,s:1,p:1
Convolution	#maps:256,k:3 × 3,s:1,p:1
MaxPooling	Window:2 × 2, s:2
Convolution	#maps:128,k:3 × 3,s:1,p:1
MaxPooling	Window:2 × 2, s:2
Convolution	#maps:64,k:3 × 3,s:1, p:1
Input	W × 32 gray-scale image

**Figure 3: The main structure of CRNN**

target detection module uses ResNet-50 model for feature pyramid type feature extraction of real scene water meter images within the dataset, then uses the DBnet algorithm to predict the probability map, threshold map and approximate binary map. The polygon of the reading area is segmented by image post-processing, and then the polygon is correctively transformed into a rectangular reading area frame, the rectangular reading area frame is input into the CRNN model for text recognition. Based on a bidirectional

LSTM decoder, character probabilities are outputted. After passing through a CTC translation layer, the predicted reading results are obtained. Finally, the predicted reading results are mapped onto each water meter image and recorded in a text file.

4 ANALYSIS OF EXPERIMENTAL RESULTS

4.1 Data Set

In this paper, we construct a dataset consisting of water meter images collected by cameras in real scenarios from several water-works plants. The dataset includes 1,500 water meter dial images of different models, different readings, different angles, and different environments, of which more than one thousand are used as the training set, and the remaining four hundred images are used as the test dataset. During the acquisition process, the coordinates of the vertices of the four corners of the water meter reading area are used as the labeled data of the training set, and the last data of the label corresponds to the correct reading of the water meter, which is used as the sample for model training.

Before the deployment of this system, the data set is first enhanced, the data enhancement in this paper mainly uses the imgaug library, which is a feature-rich and powerful image data enhancement library that can meet most of the needs of data enhancement. The data enhancement used in this paper mainly includes Random-CropData, flip, rotate and size transformation of BaseAugment, and saturation, brightness, contrast transformation of ColorJitter. The enhanced data richness improves the recognition accuracy of the model to some extent.

4.2 Improved Methods for Half-character Recognition

In the early stage of the experiment, we found that the accuracy of reading recognition is 62% when the water meter reading characters

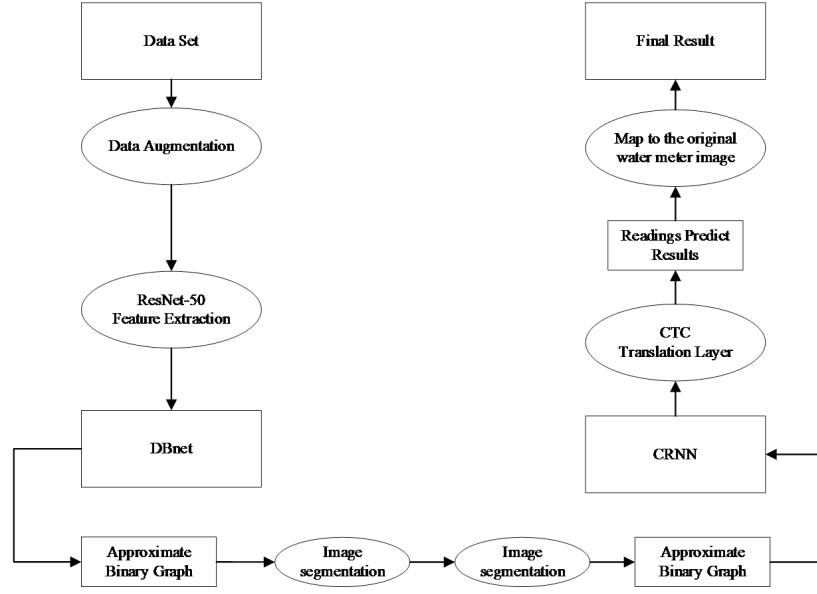


Figure 4: Framework Design

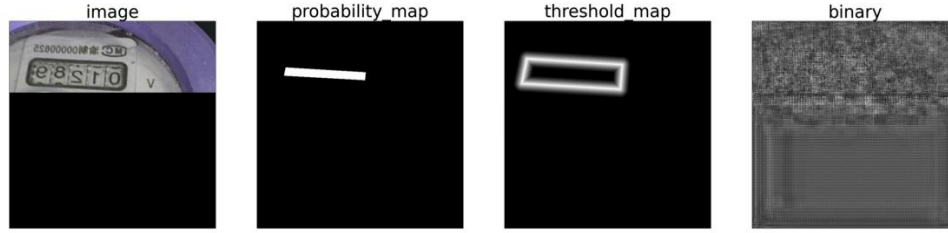


Figure 5: probability map, threshold map, and approximate binary map of DBnet

are in the "intermediate state", which is at a low level, by sampling 100 photos of water meters in real scenarios. In the middle of the experiment, two ways are proposed to increase the accuracy of reading recognition in the "intermediate state":

For the half character in the "intermediate state", it finely re-tag in the data set stage, and one half character corresponds to two reads to identify the labeled data. Ten "half characters" are learned as ten new characters by neural network, and finally mapped to numbers one by one. In the training phase, each half character data is mapped to a new symbol, the half character between 0 and 1 is mapped to A, the half character between 1 and 2 is mapped to B etc. Since according to the traditional water meter reading, the data between two numbers is considered as the smaller result. Therefore, in the post-processing stage, letters A-J need to be mapped back to numbers 0-9. According to the above half character recognition improvement method, the accuracy of half character recognition in the "intermediate state" is improved from 61% in the early stage of the experiment to 81% in the final stage.

4.3 Experiment Results and Analysis

Figure 5 shows the probability map, threshold map, and approximate binary map of DBnet output.

Figure 6 illustrates the region of reading recognition dependent on the output of DBnet for approximate binary image localization

In order to test the accuracy and speed of the water meter reading recognition system using differential binary network (DBnet) algorithm and CRNN text recognition model in real-world scenarios more accurately, this paper uses about five hundred test set images in the dataset to test the data in three aspects: the accuracy of the water meter reading region detection, the total accuracy of the reading recognition, and the accuracy of reading recognition with characters in the "in-between state".

According to the final experimental test data, in the five hundred real scene water meter reading recognition tests under different shooting angles and different lighting environments, the comprehensive recognition accuracy reached 89%, and the average recognition speed was about 0.3s. The accuracy of water meter reading area detection reaches 98%, indicating that the differential binary network (DBnet) algorithm still performs very well in target detection under varying degrees of optical or angular interference; In

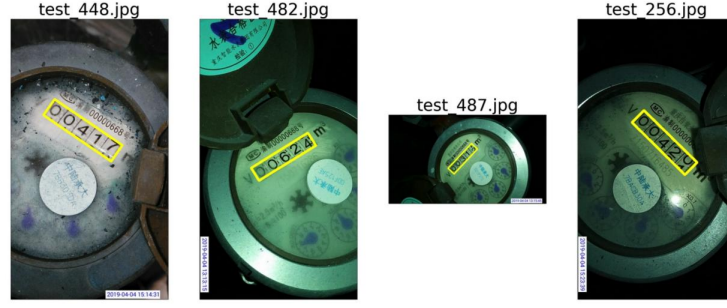


Figure 6: Positioning of the reading areas

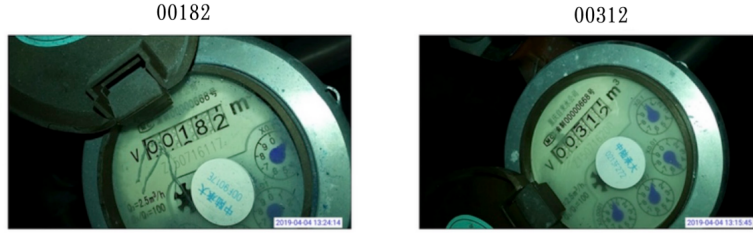


Figure 7: Correct reading identification case 1



Figure 8: Correct reading identification case 2

the clear state of reading water meter digits, the overall recognition accuracy reached 92%, demonstrating that CRNN can efficiently recognize water meter text characters with high efficiency. Under the 'intermediate state' of reading numerical symbols on the water meter, the overall recognition accuracy after improvement is approximately 81%. While there has been a significant improvement, it is still at a relatively low level. This indicates that CRNN has limited generalization ability when recognizing incomplete characters or with other characters interfering. According to the sampling statistics of error recognition results, it was found that there were significant errors in the recognition of 3 and 4, 7 and 8 half-characters by the system.

5 CONCLUSION

This paper proposes a reading recognition method applicable to water meters, which uses the ResNet-50 model for feature pyramidal type feature extraction of real scene water meter images within the dataset, and then predicts the approximate binary map through Differentiable Binarized network(DBnet), segments the

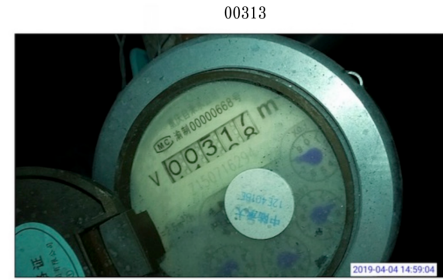


Figure 9: False reading identification case

polygon of the reading region through image post-processing, then corrects the polygon and transforms it to a rectangular reading region frame, inputs the rectangular reading region frame into the CRNN model of text recognition, and then the character probability is outputted by the bi-directional LSTM-based coder. Finally, the predicted reading result is obtained through a CTC translation

layer. The practicality of the character reading recognition system is demonstrated through the analysis of the recognition test data on the test set images

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